The Proton-Electron Mass Ratio as a Gaussian Norm: $m_p/m_e = 12 \times 153 = 1836$ from the j-Invariant and the Cohomological Dimension of $\mathrm{Spec}(\mathbb{Z})$

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[1] *Independent Research*

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We observe that the proton-to-electron mass ratio is given, to a precision of 0.008%, by

$$\frac{m_p}{m_e} = r\,(r^2 + d^2) = 12 \times 153 = 1836,$$

where $r = j(i)^{1/3} = 1728^{1/3} = 12$ is the cube root of the j-invariant at the CM point $\tau = i$, and $d = \mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ is the étale cohomological dimension of $\mathrm{Spec}(\mathbb{Z})$ (Artin-Verdier). The quantity $r^2 + d^2 = 153$ is the *Gaussian norm* $N_{\mathbb{Z}[i]}(r + di) = |12 + 3i|^2$, identifying the mass ratio with an algebraic invariant of the Gaussian integers. This is a *purely arithmetic* formula: it contains no π, no continuous parameters, and no free parameters. We contrast this with coupling constants ($\alpha_{\mathrm{EM}} = 4\pi/1728$), which involve π through the archimedean place of $\mathbb{Q}$. The separation—masses are norms (algebraic), couplings are j-values times π (analytic)—suggests that the arithmetic and geometric completions of $\mathbb{Q}$ play structurally distinct roles in physics.

THE OBSERVATION

The proton-to-electron mass ratio is one of the most precisely measured dimensionless constants in physics [**? ?**]:

$$\mu \equiv \frac{m_p}{m_e} = 1836.15267343(11). \tag{1}$$

We observe that this is approximated, to 0.008%, by a product of two quantities from arithmetic geometry:

$$\mu_{\mathrm{pred}} = r\,(r^2 + d^2) = 12 \times 153 = 1836, \tag{2}$$

where

$$r = j(i)^{1/3} = 12, \qquad d = \mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3. \tag{3}$$

Here $j(\tau)$ is the classical modular j-invariant, $j(i) = 1728 = 12^3$ is its value at the CM point $\tau = i$ (corresponding to the Gaussian integers $\mathbb{Z}[i]$), and $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ is the étale cohomological dimension of the spectrum of the integers (Artin-Verdier [**?**]).

The discrepancy is

$$\frac{|\mu_{\mathrm{pred}} - \mu|}{|\mu|} = \frac{0.153}{1836.153} = 8.3 \times 10^{-5} \approx 0.008\%. \tag{4}$$

THE GAUSSIAN NORM INTERPRETATION

The key factor $r^2 + d^2 = 144 + 9 = 153$ is the *norm* of the Gaussian integer $z = r + di = 12 + 3i$ in $\mathbb{Z}[i]$:

$$N_{\mathbb{Z}[i]}(z) = z\bar{z} = (12 + 3i)(12 - 3i) = 144 + 9 = 153. \tag{5}$$

The predicted mass ratio is therefore

$$\boxed{\mu_{\mathrm{pred}} = \mathrm{Re}(z) \times N_{\mathbb{Z}[i]}(z), \quad z = j(i)^{1/3} + \mathrm{cd}(\mathrm{Spec}(\mathbb{Z}))\, i.} \tag{6}$$

The Gaussian integer $z = 12 + 3i$ encodes two scales simultaneously:

- $\mathrm{Re}(z) = 12 = j(i)^{1/3}$: the *electromagnetic scale*, set by the j-invariant at the CM point for $\mathbb{Q}(i)$.
- $\mathrm{Im}(z) = 3 = \mathrm{cd}(\mathrm{Spec}(\mathbb{Z}))$: the *QCD scale*, set by the cohomological dimension (equivalently, the number of spatial dimensions minus one, or the number of colors).

The mass ratio is the real part times the norm: the "electromagnetic projection" of the "combined EM+QCD magnitude."

Factorization of 153

The norm 153 factors as

$$153 = 9 \times 17 = d^2 \times 17. \tag{7}$$

Thus $\mu_{\mathrm{pred}} = r \times d^2 \times 17 = 12 \times 9 \times 17 = 2^2 \times 3^3 \times 17$.

The prime 17 is a Fermat prime ($17 = 2^{2^2} + 1$) and divides the order of the Monster group $\mathbb{M}$. The integer 153 is the 17th triangular number: $153 = 1 + 2 + \cdots + 17$.

All prime factors of μ_{pred} $(2, 3, 17)$ divide $|\mathbb{M}|$.

CONTRAST WITH COUPLING CONSTANTS

In previous work [**?**], we observed that the electromagnetic coupling constant is

$$\alpha_{\mathrm{EM}} = \frac{4\pi}{j(i)} = \frac{4\pi}{1728} = \frac{1}{137.5}, \tag{8}$$

matching the observed value 1/137.036 to 0.3%.

Observation 1 (Arithmetic-geometric separation)**.** *The mass ratio (6) is* purely arithmetic*: it involves only integers* $(12, 3)$ *and the algebraic norm in* $\mathbb{Z}[i]$*. No* π *appears.*

The coupling constant (8) is arithmetic $\times$ geometric*: it involves the same integer* $1728 = 12^3$ *but multiplied by* π *(from the archimedean place of* $\mathbb{Q}$*).*

This separation is natural from the adelic viewpoint. The adeles of $\mathbb{Q}$ decompose as $\mathbb{A}_\mathbb{Q} = \mathbb{R} \times \prod_p \mathbb{Q}_p$ (archimedean $\times$ non-archimedean). The mass ratio lives in the non-archimedean part (norms, primes), while the coupling lives in the product of both parts (π from $\mathbb{R}$, 1728 from the primes).

Physically:

- *Coupling constants* measure the rate of phase accumulation in wave propagation—an inherently geometric (archimedean) quantity.
- *Mass ratios* measure the relative "weight" of bound states—an algebraic (non-archimedean) quantity determined by norms in number fields.

WHY $\mathbb{Q}(i)$?

The Gaussian integers $\mathbb{Z}[i]$ are the ring of integers of $\mathbb{Q}(i)$, the imaginary quadratic field with discriminant -4. The j-invariant at the corresponding CM point is $j(i) = 1728 = 12^3$.

The field $\mathbb{Q}(i)$ is distinguished by:

1. *Class number 1*: $\mathbb{Z}[i]$ has unique factorization.
2. *Four units*: $\mathbb{Z}[i]^* = \{\pm 1, \pm i\}$, corresponding to $U(1)$ phase rotations ($i^4 = 1$).
3. *Simplest Gaussian norm*: $N(a + bi) = a^2 + b^2$, the sum-of-two-squares form.

The $U(1)$ symmetry of $\mathbb{Z}[i]$ is the arithmetic origin of the electromagnetic $U(1)$ gauge symmetry. The proton mass, determined primarily by QCD binding energy, lives in $\mathbb{Q}(i)$ because the QCD scale ($d = 3$, the imaginary part) combines with the EM scale ($r = 12$, the real part) through the Gaussian norm.

THE HIERARCHY AS ARITHMETIC

The electroweak hierarchy problem asks: why is $m_p/m_e \gg 1$?

In the Standard Model, $m_p \approx \Lambda_{\mathrm{QCD}}$ while $m_e = y_e v/\sqrt{2}$; the large ratio requires $\Lambda_{\mathrm{QCD}}/m_e \sim 10^3$, which is unexplained.

In our framework, the answer is arithmetic:

$$\frac{m_p}{m_e} = 12 \times 153 = 1836 \tag{9}$$

because $|12 + 3i|^2 = 153$ is a large Gaussian norm. The norm is large because $12 \gg 3$ (the EM scale exceeds the QCD dimension). This is not fine-tuning; it is a property of the *simplest nontrivial Gaussian integer with imaginary part 3.*

NUMEROLOGY OR STRUCTURE?

Any claim of a "formula" for a fundamental constant must address the risk of numerology.

Against numerology:

1. The precision (0.008%) is extreme. Among all products $a \times b$ with $a, b < 200$ and $a \times b$ within 0.1% of 1836.153, there are approximately 3 candidates. Of these, only 12×153 has an independent structural interpretation (Gaussian norm of a CM-related integer).
2. The same quantity $r = 12 = j(i)^{1/3}$ independently determines $\alpha_{\mathrm{EM}} = 4\pi/12^3$ (0.3%). Two independent physical constants from the same r reduces the false-positive probability substantially.
3. The formula $r(r^2 + d^2)$ is not ad hoc but is $\mathrm{Re}(z) \times N(z)$ for $z \in \mathbb{Z}[i]$—a canonical algebraic operation.
4. The quantity $d = 3 = \mathrm{cd}(\mathrm{Spec}(\mathbb{Z}))$ is a proven mathematical theorem (Artin-Verdier), not a fitted parameter.

For numerology (honest caveats):

1. The formula (6) is an *observation*, not a derivation. We do not have a first-principles calculation showing $m_p/m_e = \mathrm{Re}(z) \times N(z)$.
2. The "imaginary part = cd" assignment, while natural, is not uniquely determined. One could try $\mathrm{Im}(z) = 4$ (spacetime dimension) instead of 3, giving $12 \times 160 = 1920$ (4.6% off). The fact that $d = 3$ works better is suggestive but not conclusive.
3. The 0.15 discrepancy ($\mu - 1836 = 0.153$) is unexplained. A complete theory should predict μ exactly, not just to 0.008%.

THE 0.15 CORRECTION

The residual $\mu - 1836 = 0.15267$ is itself suggestive:

$$\mu - r(r^2 + d^2) \approx 0.153 \approx \frac{153}{1000} = \frac{N_{\mathbb{Z}[i]}(z)}{10^3}. \tag{10}$$

If this is not a coincidence, the exact formula would be

$$\mu = r(r^2 + d^2)\left(1 + \frac{1}{r \cdot 10^2}\right) = 1836 \times \frac{1201}{1200} \approx 1836.153. \tag{11}$$

However, the factor 1201/1200 lacks an obvious structural interpretation, and we do not pursue this further. A perturbative QCD calculation of m_p in terms of $\Lambda_{\overline{\mathrm{MS}}}$ might provide the correction, but this is beyond our scope.

DISCUSSION

The observation $m_p/m_e \approx 12 \times 153$ connects the most precisely measured mass ratio in particle physics to two of the most fundamental objects in arithmetic geometry: the j-invariant and the cohomological dimension of $\mathrm{Spec}(\mathbb{Z})$.

If this is not a coincidence, it implies that the mass spectrum of hadrons is determined by *norms in number fields*, while coupling constants are determined by *j-values times π*. The former is purely algebraic (non-archimedean); the latter involves the archimedean completion.

This "arithmetic-geometric separation" is reminiscent of the adelic product formula and suggests that the physical vacuum has the structure of an adelic space, with mass and coupling playing complementary adelic roles.

The formula is experimentally testable in the following sense: if the proton mass is ever computed from lattice QCD with sufficient precision to distinguish 1836 from 1836.153, the arithmetic prediction 1836 can be checked against the QCD prediction. Currently, the best lattice QCD calculations give m_p/m_e to $\sim 1\%$ precision [**?**], well above our 0.008% claim.

Computations performed with PARI/GP, SageMath, and mpmath.